

1 Basic Category Theory

Remark 1.1. The notion of “class” in ZFC requires a definition. This can be done (“parametrically definable classes”— e.g., talk about “sets satisfying a formula” as opposed to “a set of things satisfying said formula”), but for now let’s blackbox the concept and just say “classes are a generalization of sets which are allowed to be ‘large’”, e.g. so a “class of all sets” makes sense even though a “set of all sets” does not. For this course, this shouldn’t matter ultimately anyways.¹

Definition 1.1. A **Category**, $\mathcal{C}$, is a triple of classes, $\text{obj}(\mathcal{C})$, $\text{Hom}(\mathcal{C})$, and “ $\circ$ ”, AKA $\text{Comp}(\mathcal{C})$, such that:

- For every $a, b \in \text{obj}(\mathcal{C})$, there is a set $\text{Hom}_{\mathcal{C}}(a, b)$ of elements of $\text{Hom}(\mathcal{C})$.
- For every $a, b, c \in \text{obj}(\mathcal{C})$, composition is a “map” $\circ : \text{Hom}_{\mathcal{C}}(a, b) \times \text{Hom}_{\mathcal{C}}(b, c) \rightarrow \text{Hom}_{\mathcal{C}}(a, c)$
 - Composition is associative, in that $(f \circ g) \circ h = f \circ (g \circ h)$ for any morphisms $f, g, h \in \text{Hom}(\mathcal{C})$ (whenever this composition makes sense).
- For every $a \in \text{obj}(\mathcal{C})$, there is an element $\text{id}_a \in \text{Hom}_{\mathcal{C}}(a, a)$ which is an identity for composition.

In short: we have objects, arrows between objects, a way of composing arrows, and identity arrows on each object. Oftentimes, when working with a category of modules or vector spaces, we will omit the name of the full category and simply denote $\text{Hom}_R(M, N)$ for (say) $\text{Hom}_{R\text{Mod}}(M, N)$.

The next definition is often of little intrinsic import², but will be a convenient formal thing to have for discussing “contravariant functors”, and more generally for giving a nice notion of “duality” for many arguments.

Definition 1.2. Given a category $\mathcal{C}$, we define its opposite category $\mathcal{C}^{\text{op}}$ to be the category with the same objects, but such that for every morphism $f : a \rightarrow b$ in $\text{Hom}(\mathcal{C})$ we have an arrow $f^{\text{op}} : b \rightarrow a$ in $\text{Hom}(\mathcal{C}^{\text{op}})$. Composition is defined by saying that, for every

$$a \xrightarrow{f} b \xrightarrow{g} c$$

in $\text{Hom}(\mathcal{C})$,

$$c \xrightarrow{(g \circ f)^{\text{op}}} a := c \xrightarrow{g^{\text{op}}} b \xrightarrow{f^{\text{op}}} a$$

in $\text{Hom}(\mathcal{C}^{\text{op}})$.

Definition 1.3. Given two categories, $\mathcal{C}$ and $\mathcal{D}$, a **functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a map on objects $F_{\text{obj}} : \text{obj}(\mathcal{C}) \rightarrow \text{obj}(\mathcal{D})$ together with a collection of maps on morphisms $F_{\text{Hom}(a,b)} : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(Fa, Fb)$ such that:

- F respects composition, e.g. $F_{\text{Hom}(b,c)}(\varphi) \circ F_{\text{Hom}(a,b)}(\psi) = F_{\text{Hom}(a,c)}(\varphi \circ \psi)$
- F maps identities to identities, e.g. $F_{\text{Hom}(a,a)}(\text{id}_a) = \text{id}_{F_{\text{obj}}(a)}$

. By convention we omit the subscripts and, for any $\mathcal{C}$ -object a or any $\mathcal{C}$ -morphism φ , simply denote Fa and $F\varphi$ for the corresponding object/morphism in $\mathcal{D}$.

You’ll note: this is basically the definition of a group homomorphism, but we append “takes identities to identities” since this isn’t a guarantee in the absense of inverses (which, in general, is just the definition of a “monoid homomorphism”— a map of groups with identity but without inverses). This is not mere coincidence, and is in some ways how you “should” think about it.³

¹This starts to actually be worth noting [especially supposing you care about working in ZFC or even NBG without universes] once certain constructions/objects (e.g. derived categories, functor categories, stacks etc.) become directly relevant to you. That said, if you aren’t a logician then I believe it *mostly* fine to blackbox such foundational issues until it comes up naturally.

²This is slightly illegitimate— there are indeed cases where a category of interest has, as its dual, another category of interest (up to equivalence). For algebraic geometers, one has the category of affine schemes being equivalent to the opposite of CRing , say— this leads to many useful connections between geometric and algebraic concepts.

³if you’d like, you can say categories are precisely “monoidoids”, and functors are “monoidoid homomorphisms”, and I *do* like because “monoidoid” is a silly fun word and I like saying it

Now, sometimes we wish to talk about things which don't *respect* composition, but instead reverse it. These still get to be called functors, but we call those “**contravariant functors**” and (even if we *really*) think of them as “composition reversing functors $\mathcal{C} \rightarrow \mathcal{D}$ ” it'll end up being nice to think of these quite formally as functors $\mathcal{C}^{op} \rightarrow \mathcal{D}$ —e.g. they “preserve reverse composition of arrows”.

Now we know what a functor is, we should talk about some examples and basic properties thereof.

Definition 1.4. A functor, $F : \mathcal{C} \rightarrow \mathcal{D}$, is said to be:

- **faithful** if the induced map $\text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(Fa, Fb)$ is injective for all $a, b \in \text{obj}(\mathcal{C})$
- **full** if the induced map $\text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(Fa, Fb)$ is surjective for all $a, b \in \text{obj}(\mathcal{C})$
- **bijective** if the induced map $\text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(Fa, Fb)$ is bijective for all $a, b \in \text{obj}(\mathcal{C})$

Now, in many contexts we want to compare two categories, talk about whether or not they're the “same”. In such circumstances we *could* require that we have a pair of inverse functors— an “isomorphism of categories”— but the problem is that this is way too strong (to the point of coming down to “how strict are we about what sets we use to define our objects”). It is, for instance, perfectly valid linear algebra-wise to imagine all finite $\mathbb{R}$ -vector spaces as just “vector spaces $\mathbb{R}^n$ ”, but the two categories *aren't* isomorphic because one only has a $\mathbb{Z}^+$ worth of sets while the other has way more. The notion of “isomorphism” is thus the “wrong definition”. For a better definition, rather than being “bijective on objects”, we require a sort of weaker notion:

Definition 1.5. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is said to be **essentially surjective** if, for every $d \in \text{obj}(\mathcal{D})$, there exists some $c \in \text{obj}(\mathcal{C})$ such that $Fc \cong d$.

That is to say, F “reaches all of $\mathcal{D}$ (up to abstract isomorphism)” With this in mind, we can formulate the following:

Definition 1.6. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is said to be a **(weak) equivalence of categories** if F is fully faithful and essentially surjective.

This is to say: morphisms in $\mathcal{C}$ and $\mathcal{D}$ can be thought of interchangeably—so long as you're willing to transport along abstract isomorphisms in $\mathcal{D}$. Subscribing to the philosophy that “objects are but vessels for morphisms” (hey, I don't wanna bother with the yoneda lemma here), this tells us the two categories are “morally” the same. The reason for the parenthetical “weak” is more or less a “you can choose to be careful about the axiom of choice”— in particular, there's a natural notion of a “(homotopy) equivalence of categories”— cf. 1.9, which is nice and cool but requires a bit more terminology. With Axiom of choice these are the same, but without axiom of choice this notion is “weaker”. Jason has said he often means “weak equivalence” when he says equivalence though (even though he's distinguished them in the past?), so mostly it's a “have the homotopic notion in mind morally, and the weak notion in mind for like 80% of practical examples.”

Definition 1.7. Given two functors, $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a **natural transformation** $\eta_{\bullet} : F \rightarrow G$ is a collection of maps for each $a \in \text{obj}(\mathcal{C})$, $\eta_a : Fa \rightarrow Ga$ such that for every $a \xrightarrow{\varphi} b$ in $\text{Hom}(\mathcal{C})$ the following square commutes:

$$\begin{array}{ccc} Fa & \xrightarrow{F\varphi} & Fb \\ \eta_a \downarrow & & \downarrow \eta_b \\ Ga & \xrightarrow{G\varphi} & Gb \end{array}$$

Remark 1.2. If the notation was not itself suggestive enough (and perhaps it wasn't at this high a level of abstraction, alas), a “natural transformation” is a good concept of “map between functors”— this should already be a somewhat familiar notion to those who know some representation theory: A representation of G over k (take $k = \mathbb{R}, \mathbb{C}$ for comfort if you like) is a sort of map (indeed, literally a functor) $G \rightarrow \text{Vect}_k$, and an equivariant map between two representations is analogous to (indeed, literally is) a natural transformation between them. People also sometimes like to illustrate these with a “globular diagram”

$$\begin{array}{ccc} & F & \\ \mathcal{C} & \searrow \eta_{\bullet} & \mathcal{D} \\ & G & \end{array}$$

, which in particular should make you imagine a sort of “directed” homotopy from F to G (and indeed, one can kind of formulate homotopies of maps of topological spaces in this “maps between maps” language blabla)– in a bit, we’ll see some concepts which are roughly “homotopy equivalences of categories”, so this intuition is by no means off the mark.

Example 1.1. Denote $(-)^{\times} : \mathbf{CRing} \rightarrow \mathbf{Group}$ the functor that associates any commutative ring R to its group of units. Denote $GL_n(-) : \mathbf{CRing} \rightarrow \mathbf{Group}$ the functor that associates any commutative ring R to the group of invertible $n \times n$ matrices with coefficients in R . Then, the determinant map defines a natural transformation $\det_{\bullet} : GL_n(-) \rightarrow (-)^{\times}$.

Example 1.2. Denote $(-)^{\vee} : {}_{\mathbb{R}}\mathbf{FinVect} \rightarrow {}_{\mathbb{R}}\mathbf{FinVect}$ the functor that takes a vector space, V , to its dual $V^{\vee} := \text{Hom}_{\mathbb{R}}(V, \mathbb{R})$. Now, for each finite dimensional vector space V , one can choose an isomorphism $\eta_V : V \xrightarrow{\sim} V^{\vee}$ by choosing a basis. Nonetheless, this can not be assembled into a natural transformation $\text{id}_{\mathbf{C}} \rightarrow (-)^{\vee}$ – intuitively because there is no way to make the choice of isomorphism “naturally” or “canonically”

“why do we care about this?” well, to curry your question into a different question, we define another thing:

Definition 1.8. A pair of functors $L : \mathcal{C} \rightarrow \mathcal{D}$, $R : \mathcal{D} \rightarrow \mathcal{C}$ are said to be an **adjoint pair** if any of the following equivalent definitions hold:

- (Hom bijection) For every $c \in \text{obj}(\mathcal{C})$, $d \in \text{obj}(\mathcal{D})$, there is a natural isomorphism $\eta_{c,d} : \text{Hom}_{\mathcal{D}}(Lc, d) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(c, Rd)$, where “natural” is taken to mean “for any $c \xrightarrow{\varphi} c' \in \text{Hom}(\mathcal{C})$, $d \xrightarrow{\psi} d' \in \text{Hom}(\mathcal{D})$, the following commutes:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(Lc, d) & \xrightarrow{\eta_{c,d}} & \text{Hom}_{\mathcal{D}}(c, Rd) \\ \text{Hom}_{\mathcal{C}}(F\varphi, \psi) \downarrow & & \downarrow \text{Hom}_{\mathcal{D}}(\varphi, R\psi) \\ \text{Hom}_{\mathcal{C}}(Lc', d') & \xrightarrow{\eta_{c',d'}} & \text{Hom}_{\mathcal{D}}(c', Rd') \end{array}$$

- (equational) We have natural transformations $\eta : \text{id}_{\mathcal{C}} \rightarrow R \circ L$ (the “unit”) and $\varepsilon : L \circ R \rightarrow \text{id}_{\mathcal{D}}$ (the “counit”) such that the following commutes:

$$\begin{array}{ccc} R & \xrightarrow[\eta_R(-)]{\text{id}_R} & RLR \xrightarrow{R\varepsilon(-)} R \\ L & \xrightarrow[\text{id}_L]{L\eta(-)} & LRL \xrightarrow{\varepsilon_L(-)} L \end{array}$$

(in a quip, “ $\varepsilon \circ \eta = \text{id}$ ”).

- (universal arrow for left adjoint) For every $c \in \mathcal{C}$, there exists $Rc \in \text{obj}(\mathcal{D})$ and a map $\varepsilon_c : L Rc \rightarrow c$ such that, for every $d \in \text{obj}(\mathcal{D})$ and every $\varphi : Ld \rightarrow c$, there exists unique $\tilde{\varphi} : d \rightarrow Rc$ making the following diagram commute

$$\begin{array}{ccc} Ld & & \\ \downarrow \exists! L\tilde{\varphi} & \searrow \varphi & \\ L Rc & \xrightarrow{\varepsilon_c} & c \end{array}$$

- (universal arrow for right adjoint) For every $d \in \mathcal{D}$, there exists $Ld \in \text{obj}(\mathcal{C})$ and a map $\eta_d : d \rightarrow R Ld$ such that, for every $c \in \text{obj}(\mathcal{C})$ and every $\psi : d \rightarrow Rc$, there exists unique $\tilde{\psi} : Ld \rightarrow c$ making the

following diagram commute

$$\begin{array}{ccc}
 & Rc & \\
 \exists ! R\tilde{\psi} \uparrow & \swarrow \psi & \\
 RLd & \xleftarrow{\eta_d} & d
 \end{array}$$

In any of these cases, L is said to be a **left adjoint** (to R), and R is said to be a **right adjoint** (to L).

Remark 1.3. Each of these definitions gives a slightly different insight into what a “adjunction” really is, namely:

Remark 1.4. Technically, a proof that these definitions are equivalent is required. Practically, it’s pure abstract nonsense and not particularly insightful. To see the proofs, look at the nlab pages on “adjunction” and “adjoint functors”, but you shouldn’t need to remember them nearly so much as (the intuition behind) the definitions.

Now that we have the requisite language, we can finally formulate a “(homotopy) equivalence of categories”:

Definition 1.9. Two categories, $\mathcal{C}, \mathcal{D}$ are said to be **equivalent** if there exists pair of functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ and a pair of natural isomorphisms $\eta : \text{id}_{\mathcal{C}} \rightarrow F \circ G$ and $\varepsilon : G \circ F \text{id}_{\mathcal{D}}$.

Remark 1.5. Any equivalence is an adjunction by the counit-unit definition above, and indeed: an equivalence is precisely an adjunction for which the counit and unit transformations are iso.

1.1 Yoneda Lemma

Definition 1.10. Let $\mathcal{C}, \mathcal{D}$ be categories. Then, we denote $\text{Fun}(\mathcal{C}, \mathcal{D})$ the category of (covariant) functors from $\mathcal{C} \rightarrow \mathcal{D}$, where the morphisms are natural transformations.

The above definition actually *doesn’t* always “make sense” in that we required categories to have a *set* of morphisms between any two objects, and yet this definition (at least taken as literally as possible) does not always meet this criterion. We shall ignore such issues for now.

Definition 1.11. Let $\mathcal{C}$ be a category. We denote by $\mathfrak{J}_{\bullet} : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}, \text{Set})$ the functor $c \mapsto \mathfrak{J}_c := \text{Hom}_{\mathcal{C}}(c, -)$, and we denote $\mathfrak{J}^{\bullet} : \mathcal{C}^{\text{op}} \rightarrow \text{Fun}(\mathcal{C}, \text{Set})$ the functor $c \mapsto \mathfrak{J}^c := \text{Hom}_{\mathcal{C}}(-, c)$. These are called the **covariant and contravariant Yoneda functors** respectively (with the symbol used being the hiragana character “yo” for Yoneda). The notation $h_{\bullet}$ or $h^{\bullet}$ is also used.

Theorem 1.1. (Yoneda Lemma) There is a natural isomorphism between the functors

$$\begin{aligned}
 \text{eval} : \mathcal{C} \times \text{Fun}(\mathcal{C}, \text{Set}) &\rightarrow \text{Set} \\
 (c, F) &\mapsto Fc \\
 (\varphi : c \rightarrow c', \eta : F \rightarrow G) &\mapsto \eta_{c'} \circ F\varphi
 \end{aligned}$$

and

$$\begin{aligned}
 \text{Hom}_{\text{Fun}(\mathcal{C}, \text{Set})}(\mathfrak{J}_{\bullet}, -) : \mathcal{C} \times \text{Fun}(\mathcal{C}, \text{Set}) &\rightarrow \text{Set} \\
 (c, F) &\mapsto \text{Hom}_{\text{Fun}(\mathcal{C}, \text{Set})}(\mathfrak{J}_c, F) \\
 (\varphi : c \rightarrow c', \eta : F \rightarrow G) &\mapsto \text{Hom}_{\text{Fun}(\mathcal{C}, \text{Set})}(\mathfrak{J}_{\varphi}, \eta).
 \end{aligned}$$

(as well as a dual one for the contravariant yoneda map). This is to say, we have natural bijections

$$\text{Hom}_{\text{Fun}(\mathcal{C}, \text{Set})}(\mathfrak{J}_c, F) \cong Fc$$

and

$$\text{Hom}_{\text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})}(\mathfrak{J}^c, F) \cong Fc.$$

Proof. For all $\varphi : c \rightarrow c', F : \mathcal{C} \rightarrow \mathbf{Set}$, we have a natural transformation $\vartheta : \mathfrak{J}_c \rightarrow F$ given by the commutative diagram

$$\begin{array}{ccc}
 \mathfrak{J}_c(c) = \mathrm{Hom}_{\mathcal{C}}(c, c) & \xrightarrow{\mathrm{Hom}_{\mathcal{C}}(c, \varphi) =: \mathfrak{J}_c \varphi} & \mathfrak{J}_c(c') = \mathrm{Hom}_{\mathcal{C}}(c, c') \\
 \downarrow \vartheta_c & \begin{array}{ccc} \mathrm{id}_c & \xrightarrow{\quad} & \varphi \\ \downarrow & & \downarrow \\ * & \xrightarrow{\quad} & F\varphi(*) = \vartheta_{c'} \varphi \end{array} & \downarrow \vartheta_{c'} \\
 Fc & \xrightarrow{F\varphi} & Fc'
 \end{array}$$

Any natural transformation ϑ is determined by the choice of $\vartheta_c(\mathrm{id}_c) = * \in F(c)$, and any choice of $*$ defines such a natural transformation.

An analogous diagram proves the analogous result for $\mathfrak{J}^\bullet$. □

Alright, so that was some very abstract nonsense. Key upshot (for now):

Corollary 1.1. *Two $\mathcal{C}$ -objects, c, c' , are isomorphic if and only if their Hom functors (contravariant or covariant) are isomorphic.*

Proof. By the above, we have

$$\mathrm{Hom}_{\mathrm{Fun}(\mathcal{C}, \mathbf{Set})}(\mathfrak{J}_c, \mathfrak{J}_{c'}) \cong \mathrm{Hom}_{\mathcal{C}}(c', c)$$

and

$$\mathrm{Hom}_{\mathrm{Fun}(\mathcal{C}^{op}, \mathbf{Set})}(\mathfrak{J}^c, \mathfrak{J}^{c'}) \cong \mathrm{Hom}_{\mathcal{C}}(c, c').$$

□

Remark 1.6. *Because of the above result, the functors $\mathfrak{J}^\bullet$ and $\mathfrak{J}_\bullet$ are often called the “(resp. contravariant, covariant) Yoneda embedding”, as we see that these functors are fully faithful (and isomorphism reflecting).⁴*

Philosophically, here’s the point: “objects in $\mathcal{C}$ are determined by morphisms to (and/or from) them”. This is, more or less, a far far reaching generalization of Cayley’s theorem in group theory (“every group encodes the symmetries of some object”).

1.2 (Co)Limits

Definition 1.12. *In a category, $\mathcal{C}$, a “**diagram**” is a functor $F : \mathcal{I} \rightarrow \mathcal{C}$ where $\mathcal{I}$ is small (and thought of as a “index category”).*

The term “index category” here is taken in analogy to that of “index set”, e.g. for a product/direct sum.

Example 1.3. *Take $\mathcal{I}$ to be the category given by the following digraph (with identity maps implicit):*

$$1 \xleftarrow{i} 3 \xrightarrow{j} 2$$

Take $\mathcal{C} = \mathbf{CRing}$. Then, the functor such that $F(1) = \mathbb{Z}[x]$, $F(2) = \mathbb{Z}/\mathbb{Z}$, $F(3) = \mathbb{Z}$ with $F(i), F(j)$ taken to be the natural inclusions gives a diagram, visualized as

$$\mathbb{Z}[x] \longleftarrow \mathbb{Z} \xrightarrow{j} \mathbb{Z}/2\mathbb{Z}$$

⁴Sometimes, it is considered useful to view certain nice contravariant functors from certain nice (“geometric/topological”) categories as “generalized spaces” (“sheaves and/or algebraic spaces”) which arise from gluing the objects/spaces in the base category. In this view, a “sheaf” which is isomorphic to $\mathfrak{J}^c$ is said to be “represented by c ” – such matters are quite relevant to the study of moduli spaces/moduli problems.

Example 1.4. Take $\mathcal{I}$ to be the category given by the following infinite digraph (with identity maps implicit):

$$1 \xrightarrow{i_{12}} 2 \xrightarrow{i_{23}} 3 \xrightarrow{i_{34}} \dots$$

Take $\mathcal{C} = \mathbf{CRing}$. Then, the functor such that, for every positive integer n , we take $F(n) = \mathbb{Z}/2^n\mathbb{Z}$ and $F(i_{n(n+1)})$ is taken to be the natural inclusion $\mathbb{Z}/2^n\mathbb{Z} \rightarrow \mathbb{Z}/2^{n+1}\mathbb{Z}$ is a diagram, visualized as

$$\mathbb{Z}/2\mathbb{Z} \longrightarrow \mathbb{Z}/2^2\mathbb{Z} \longrightarrow \mathbb{Z}/2^3\mathbb{Z} \longrightarrow \dots$$

Example 1.5. Take $\mathcal{I}$ to be the category given by the following digraph (with identity maps implicit):

$$1 \quad 2$$

Take $\mathcal{C} = \mathbf{CRing}$. Then, the functor with $F(1) = \mathbb{R}$ and $F(2) = \mathbb{C}$ gives a diagram, visualized as:

$$\mathbb{R} \quad \mathbb{C}$$

Definition 1.13. Let $F : \mathcal{I} \rightarrow \mathcal{C}$ be a diagram. A “**cone over**” (resp. “**cocone under**”) is a $\mathcal{C}$ -object c , together with arrows $c \xrightarrow{\pi_n} Fn$ (resp. $c \xleftarrow{\iota_n} Fn$) for each $n \in \text{obj}(\mathcal{I})$, such that for every $n \xrightarrow{i_{nm}} m \in \text{Hom}(\mathcal{I})$ we have that the following diagram on the left (resp. on the right) commutes:

$$\begin{array}{ccc} & c & \\ \pi_n \swarrow & & \searrow \pi_m \\ Fn & \xrightarrow{F(i_{nm})} & Fm \end{array} \quad \begin{array}{ccc} Fn & \xrightarrow{F(i_{nm})} & Fm \\ \iota_n \searrow & & \swarrow \iota_m \\ & c & \end{array}$$

Definition 1.14. Let $F : \mathcal{I} \rightarrow \mathcal{C}$ be a diagram. A **limit** (resp. **colimit**) is a terminal cone over F (resp. initial cocone under F)— e.g, an object c equipped with maps $c \xrightarrow{\pi_n} Fn$ (resp. $c \xleftarrow{\iota_n} Fn$) such that, for any other object c' equipped with maps $c' \xrightarrow{\pi'_n} Fn$ (resp. $c' \xleftarrow{\iota'_n} Fn$), there exists a unique map $\pi : c' \rightarrow c$ (resp. $\iota : c \rightarrow c'$) such that the following diagram on the left (resp. on the right) commutes:

$$\begin{array}{ccc} & c' & \\ \pi'_n \swarrow & \downarrow \exists! \pi & \searrow \pi'_m \\ & c & \\ \pi_n \swarrow & & \searrow \pi_m \\ Fn & \xrightarrow{F(i_{nm})} & Fm \end{array} \quad \begin{array}{ccc} Fn & \xrightarrow{F(i_{nm})} & Fm \\ \iota_n \searrow & & \swarrow \iota_m \\ & c & \\ \iota'_n \swarrow & \downarrow \exists! \iota & \searrow \iota'_m \\ & c' & \end{array}$$

Example 1.6. Let $F : \mathcal{I} \rightarrow \mathcal{C}$ be as in example 1.3. Then, the colimit over the diagram is given by the tensor product as (commutative) $\mathbb{Z}$ -algebras, $\mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}[x]$:

$$\begin{array}{ccc} \mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} & \longleftarrow & \mathbb{Z}[x] \\ \uparrow & & \uparrow \\ \mathbb{Z}/2\mathbb{Z} & \longleftarrow & \mathbb{Z} \end{array}$$

This is also called a **pushout** (its dual is called a **pullback** or a **fibre product**).

Example 1.7. Let $F : \mathcal{I} \rightarrow \mathcal{C}$ be as in example 1.4. Then, the limit over the diagram is given by the 2-adic integers $\mathbb{Z}_2$ (look them up on wikipedia if you haven't seen them before, they're useful for number theory etc.), together with the natural projections:

$$\begin{array}{ccccc} & & \mathbb{Z}_2 & & \\ & \nearrow & \uparrow & \nwarrow & \\ \mathbb{Z}/2\mathbb{Z} & \longrightarrow & \mathbb{Z}/2^2\mathbb{Z} & \longrightarrow & \mathbb{Z}/2^3\mathbb{Z} \longrightarrow \dots \end{array}$$

This is also called a **projective/inverse limit** (its dual is called a **direct/inductive limit**).

Example 1.8. Let $F : \mathcal{I} \rightarrow \mathcal{C}$ be as in example 1.5. Then, the limit over the diagram is given by the product as rings, and the colimit over the diagram is given by the tensor product as $\mathbb{Z}$ -algebras (note $\mathbb{Z}$ the initial object of $\mathbf{CRing}$, so one has “invisible” maps $\mathbb{Z} \rightarrow \mathbb{R}$ and $\mathbb{Z} \rightarrow \mathbb{C}$):

$$\begin{array}{ccc} & \mathbb{R} \times \mathbb{C} & \\ \pi_{\mathbb{R}} \swarrow & & \searrow \pi_{\mathbb{C}} \\ \mathbb{R} & & \mathbb{C} \end{array} \qquad \begin{array}{ccc} \mathbb{R} & & \mathbb{C} \\ \iota_1 \searrow & & \swarrow \iota_2 \\ & \mathbb{R} \otimes_{\mathbb{Z}} \mathbb{C} & \end{array}$$

The left limit is called a **product**, the right colimit is called a **coproduct** (though indeed, coproducts in $\mathbf{CRing}$ can be thought of as pushouts under $\mathbb{Z}$).

Idea: colimits usually look like disjoint unions modulo identifying some things, limits usually look like subcollections of products. Google examples of (co)products, pushouts, and pullbacks in categories you know about (including $\mathbf{Top}$ by the way) to feel a bit more comfortable with the terminology. Important note: in module categories, finite (but not infinite!) products and coproducts coincide. “coproducts” bear the name of “direct sum” more generally, but finite direct sums *are* products as well (which is part of what makes “abelian categories” so nice).

Theorem 1.2. Suppose $F : \mathcal{C} \rightarrow \mathcal{D}$ is a functor. Then:

- If F is a left adjoint (e.g., has a right adjoint), then F preserves colimits.
- If F is a right adjoint (e.g., has a left adjoint), then F preserves limits.

This has a proof by pure formal nonsense given by unraveling the definitions. This is also an incredibly useful fact from which many other properties follow. If it helps to remember, one has the mnemonics “LAPC [left adjoints preserve colimits]” and “RAPL [right adjoints preserve limits].”

Proposition 1.1. Let $F : \mathcal{I} \rightarrow \mathcal{C}$ be a diagram. Then, a limit of F (resp. a colimit of F) is precisely a $\mathcal{C}$ -object $\lim F$ (resp. $\operatorname{colim} F$) such that, for all $c \in \operatorname{obj}(\mathcal{C})$ we have a natural bijection

$$\operatorname{Hom}_{\mathcal{C}}(c, \lim F) \cong \operatorname{Hom}_{\operatorname{Fun}(\mathcal{I}, \mathcal{C})}(\operatorname{const}_c, F)$$

or respectively a natural bijection

$$\operatorname{Hom}_{\mathcal{C}}(\operatorname{colim} F, c) \cong \operatorname{Hom}_{\operatorname{Fun}(\mathcal{I}, \mathcal{C})}(F, \operatorname{const}_c),$$

where by $\operatorname{Fun}(\mathcal{I}, \mathcal{C})$ we mean the category of functors and natural transformations (ignore set-theoretic issues here), and by const_c we mean the functor that takes every object to c and every morphism to id_c .

Rephrasing: $\lim_{\mathcal{I}}$ and $\operatorname{colim}_{\mathcal{I}}$ are right (resp. left) adjoint to the functor $\operatorname{const} : \mathcal{C} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$.

This above proposition, modulo set-theoretic issues w.r.t. defining a functor category, is a valid alternative definition of “limit/colimit”, and can lead one to various useful ideas (e.g., the 2-category of categories, viewing $\lim / \operatorname{colim}$ as functors on their own right, left/right Kan extensions, etc.). It is a fine exercise to try and unravel this definition and see how the universal property (w.r.t. cones/cocones) is equivalent to this statement, but we would suggest not overemphasizing this definition if any of these concepts are new to the reader.

2 Tensor products (even over potentially non-commutative rings) and more miscellaneous module theory

Definition 2.1. *R*-balanced maps Let R be a ring, $M \in \text{Mod}_R, N \in {}_R\text{Mod}$. Then, an R balanced map $\varphi : M \times N \rightarrow A$ an abelian group is a biadditive map (e.g., $\mathbb{Z}$ -bilinear map) such that, for all $r \in R$, $\varphi(mr, n) = \varphi(m, rn)$ for all $m \in M, n \in N$.

Definition 2.2. *tensor product* Let $M \in {}_S\text{Mod}_R, N \in {}_R\text{Mod}_{S'}$ for a ring R . Then, the tensor product $M \otimes_R N$ is an abelian group (in particular, an S - S' -bimodule) together with a map $\otimes_R : M \times N \rightarrow M \otimes_R N$ such that, for any abelian group A together with an R -balanced map $\varphi : M \times N \rightarrow A$, there exists unique abelian group homomorphism $\tilde{\varphi} : M \otimes_R N \rightarrow A$ making the following diagram commute:

$$\begin{array}{ccc} M \times N & \xrightarrow{\otimes} & M \otimes_R N \\ & \searrow \varphi & \downarrow \exists! \tilde{\varphi} \\ & & A \end{array}$$

. If R is commutative, then $M \otimes_R N$ admits the natural structure of an R -module, and we can instead consider the universal R -module such that any R -bilinear map $M \times N \rightarrow A$ factors through an R -module homomorphism $M \otimes_R N \rightarrow A$.

Note: while we take bimodules above, we can simply take S, S' , or R to be $\mathbb{Z}$ and recover “one-sided” modules. That’s a universal property or what have you— really, it’s the same philosophy as viewing the “tensor product of V and W ” as “the multilinear maps from $V \times W$ ”, we’re just being cautious about commutativity and being a smidgeon more careful since not every module is free here. Anyways, “how/why does this universal object actually exist?”, well, we just take a super big free module and the mod out by what it means to be R -balanced, aka by what it means to be (non-commutatively) multilinear. We give the proof of existence as follows:

Lemma 2.1.)(tensor products exist)

The tensor product has, itself, a deep relation to Hom -functors— namely, we have the *tensor-hom adjunction*:

Theorem 2.1.

Definition 2.3. Let R be a ring. An R -**algebra** is a ring, S , together with a ring homomorphism $R \rightarrow S$ (the “structure map”).

Definition 2.4. Let $R \rightarrow S, R \rightarrow S'$ be two R -algebras. A **(homo)morphism of R -algebras** from $S \rightarrow S'$ is a ring homomorphism $\varphi : S \rightarrow S'$ which commutes with the structure maps, e.g. making the following diagram commute:

$$\begin{array}{ccc} & R & \\ & \swarrow \quad \searrow & \\ S & \xrightarrow{\varphi} & S' \end{array}$$

Definition 2.5. Let R be a commutative ring, let $R \rightarrow S$ and $R \rightarrow S'$ be R -algebras. Then, the tensor product of R -algebras is the colimit in CRing over the diagram

$$S' \leftarrow R \rightarrow S,$$

e.g. the pushout in CRing (or, as one can verify, the coproduct in ${}_R\text{CAlg}$).

Remark 2.1. Note that the tensor product of R -algebras can also be taken in steps. Namely, any R -algebra $R \rightarrow S$ admits the natural structure of an R -module by left/right multiplication. Then, for two R -algebras S and S' , one can take their tensor product as R -modules, and then endow $S \otimes_R S'$ with the structure of an ring by defining $(s_1 \otimes s'_1)(s_2 \otimes s'_2) := s_1 s_2 \otimes s'_1 s'_2$, which one can check gives an R -algebra in the natural way.

Definition 2.6. Let R be a commutative ring, and $S \subset R$ a multiplicatively closed subset. The **localization**, $S^{-1}R$ is taken to be a (commutative) R -algebra $R \rightarrow S^{-1}R$ such that, for any ring homomorphism $R \rightarrow T$ taking every element of S to a unit, there exists a unique ring homomorphism $S^{-1}R \rightarrow T$ making the following diagram commute:

$$\begin{array}{ccc} R & \xrightarrow{\quad} & S^{-1}R \\ & \searrow & \downarrow \exists! \\ & & T \end{array}$$

This is to say, $S^{-1}R$ is the “smallest” (e.g., initial) R -algebra for which every element of S is a unit. As a universal property, this requires a construction/proof of existence—this one justifies the common name “ring of fractions” this ring.

Theorem 2.2. (Construction of localizations) *Localizations exist.*

Proof. Let S be a multiplicative subset of R commutative. Define, on $R \times S$, an equivalence relation

$$(a, b) \sim (c, d)$$

if there exists $s \in S$ such that

$$s(ac - bd) = 0.$$

As this forms an ideal, we may take $S^{-1}R$ to be the ring of equivalence classes under this relation; we denote the equivalence class of (r, s) as either $\frac{r}{s}$ or $s^{-1}r$. Then, the map $r \mapsto \frac{r}{1} : R \rightarrow S^{-1}R$ makes $S^{-1}R$ initial among all R -algebras for which S is taken to units by the structure map. $\square$

The relation in question should be taken to be exactly that of fraction addition working “as it should”. The use of an extra $s \in S$ in that definition is mostly just to fudge things a little so that we don’t need to make an extra case for the multiplicative set having zero divisors (but, for S to be a multiplicative set containing a zero divisor, it must also contain 0 and hence the localization must be the zero ring anyways.)

There is some special notation in the following cases:

Definition 2.7. If $f \in R$, then we denote $R_f := \{f, f^2, f^3, \dots\}^{-1}R$ the “**localization away from f** ”. If $\mathfrak{p}$ is a prime ideal of R , then we denote $R_{\mathfrak{p}} := (R - \mathfrak{p})^{-1}R$ the “**localization at $\mathfrak{p}$** ”.⁵

⁵This terminology comes from algebraic geometry, where prime ideals (or, intuitively, maximal ideals) are thought of as “points” of a topological space. Imagining R as the ring of continuous functions on a compact Hausdorff space, $\mathfrak{p}$ as a point of said space, then we have the localization *at* $\mathfrak{p}$ arising as the “ring of germs of functions at $\mathfrak{p}$ ”. The terminology for localization *away from* an element is analogous: the localization at $f \in R$ is precisely the localization (intuitively, restriction to) all prime ideals (“points”) not containing f (that is, where f does not vanish).

3 (co)homological algebra

For the duration of this section, by “module category”, I will generally mean ${}_R\mathbf{Mod}_S$ for some (not-necessarily-commutative) rings R, S . Noting that we can recover a one-sided module category by taking R or S to be $\mathbb{Z}$, and that for many of these statements we don’t care about “sidedness”, we will save some trouble and just say “module category”, and denote them generically by $\mathcal{A}$ or $\mathcal{B}$.

Definition 3.1. If $F : \mathcal{A} \rightarrow \mathcal{B}$ is a functor of module categories (it can be different sided module categories too, it doesn’t matter), F is said to be *additive* if, given any two $\varphi, \psi \in \mathbf{Hom}_R(M, N)$, we have $F(\varphi + \psi) = F(\varphi) + F(\psi)$ —that is, on each $\mathbf{Hom}$ it’s a homomorphism of additive groups.

I’m really only mentioning the word “additive” here so that I can say “additive” later on without needing to refer you to the stuff on abelian categories (at least, not yet).

Now, to start with, we want to talk about “exactness.”

Definition 3.2. A sequence of R -modules/homomorphisms,

$$\cdots \rightarrow A \xrightarrow{\varphi} B \xrightarrow{\psi} C \rightarrow \cdots$$

is said to be “**exact at B** ” if $\ker \psi = \operatorname{im} \varphi$. If the sequence is exact at every module, then it is sometimes called a **long exact sequence** (abbreviated “LES”).

Definition 3.3. A sequence of R -modules of the form

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

is said to be a **short exact sequence** (abbreviated “SES”) if it is exact at A , B , and C (e.g., a LES of the above shape).

Take note that exactness at the left/right terms of a SES is precisely injectivity/surjectivity respectively. There are many natural sequences—“complexes”—of R -modules which come up in various places in mathematics; oftentimes, we wish to view failure of a sequence to be exact as being a “obstruction” to some other phenomenon. For instance, in topology, one might care to ask if the singular/simplicial (co)chain complex is exact, and the extent to which this fails is precisely due to the presence of “holes” in each dimension (and by De Rham’s theorem, the same is true for, say, De Rham cohomology on a smooth manifold)—thinking in reverse, we can gain insight about our objects of interest by measuring the failure of certain algebraic “complexes” to be exact. The goal of (co)homological algebra is principally to systematize how we study/discuss such scenarios.

Having established that exactness of sequences is a key interest, we ask the following: “When does a functor *preserve* exactness?” To help discuss and elaborate upon this, we introduce some definitions.

Definition 3.4. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor of module categories. Then, if for any short exact sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

we have

- $0 \rightarrow FA \rightarrow FB$ exact at FA , then we say F is **left exact**
- $FB \rightarrow FC \rightarrow 0$ exact at FC , then we say F is **right exact**
- $FA \rightarrow FB \rightarrow FC$ exact at FB , then we say F is **half exact**
- $0 \rightarrow FA \rightarrow FB \rightarrow FC \rightarrow 0$ a SES, then we say F is **exact**.

Note that left/right exactness is equivalent taking injections (“monos”) to injections and surjections (“epis”) to surjections respectively. In fact, one can show the following:

Proposition 3.1. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor of module categories. Then, F is left (resp. right) exact if and only if F preserves finite limits (resp. colimits).

Proof. Note the reverse direction is clear– if F preserves finite limits (resp. colimits), then it preserves kernels (resp. cokernels), hence takes any morphism with (co)kernel 0 to one with (co)kernel 0.

For the forward implication, we note some general abstract nonsense. Namely, as an additive functor, F necessarily “preserves finite (bi)products” (that is, finite products and coproducts, which happen to coincide). Secondly, since F preserves injections (resp. surjections), we can thus surmise that F preserves all kernels (resp. cokernels). Now, by general abstract nonsense, it is known that all finite (co)limits can be built out of (co)kernels and finite (co)products, and since F preserves those, F preserves *all* finite (co)limits. $\square$

The proof is not particularly instructive here (not yet, anyways), but it’s nice to know that one can take pullbacks/pushouts or kernels/cokernels etc. without needing to worry about if F preserves them. This also tells us yet more cleanly that any left adjoint is right exact, and any right adjoint is left exact. Another thing worth noting is that “half-exactness” is often not necessary to prove/worry too much about by the following result:

Proposition 3.2. *An additive functor $F : \mathcal{A} \rightarrow \mathcal{B}$ of module categories is exact if and only if it is both left and right exact.*

Proof. The forward implication is clear. The reverse implication follows by the (proof of the) previous proposition: since F is left and right exact, F preserves kernels and cokernels. $\square$

Example 3.1. *For any right R -module A , the functor $A \otimes_R -$ is right exact.*

Example 3.2. *For any R -module A , the functor $\text{Hom}_R(A, -)$ is left exact.*

Example 3.3. *For any commutative ring R , S a multiplicative subset of R , the localization functor $S^{-1}(-) : {}_R\text{Mod} \rightarrow {}_{S^{-1}R}\text{Mod}$ is exact.*

Now, we need to be a bit cautious here when we talk about *contravariant* functors like $\text{Hom}_R(-, A)$. In particular, the opposite of a module category is *not* a module category, so we either need to formulate things in terms of abelian categories, *or* we need to rewrite some definitions. For concreteness/ease of reference, we do the latter:

Definition 3.5. *Let $F : \mathcal{A}^{\text{op}} \rightarrow \mathcal{B}$ be an additive functor of module categories. Then, if for any short exact sequence*

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

we have

- $0 \rightarrow FC \rightarrow FB$ exact at FC , then we say F is **left exact**
- $FB \rightarrow FA \rightarrow 0$ exact at FA , then we say F is **right exact**
- $FC \rightarrow FB \rightarrow FA$ exact at FB , then we say F is **half exact**
- $0 \rightarrow FC \rightarrow FB \rightarrow FA \rightarrow 0$ a SES, then we say F is **exact**.

This gives us yet another example:

Example 3.4. *For any R -module A , the functor $\text{Hom}_R(-, A)$ is left exact.*

Now, we have various examples of additive functors in hand which satisfy various exactness properties, *but* the vast majority of them are *not exact*. This gives us another question, namely: “Is there some way to ‘fix’ left-exact or right-exact functors and ‘turn them exact’?” It turns out that the answer will be yes: in module categories, we can “derive” these left/right-exact functors in a nice way. As (co)homology theories in the topological or smooth categories measure “obstructions to exactness”, so too will we define “(co)homology theories” in module categories– that is, Ext and Tor – which measure obstructions to certain functors being exact. Broadly speaking, the way we do this will involve taking the “not-quite-exact sequence” obtained by applying the functor, and then “resolving” the sequence to get along exact sequence. To understand where these resolutions come from, we need to define certain classes of “nice” modules which we can make a resolution out of.

Definition 3.6. An R module, P , is said to be **projective** if any of the following equivalent statements are true:

- For any surjective module homomorphism $N \twoheadrightarrow M$, any module homomorphism $P \rightarrow M$ admits a (not-necessarily unique) lift $P \rightarrow N$ such that the following diagram commutes:

$$\begin{array}{ccc} & & N \\ & \nearrow \text{dashed} & \downarrow \\ P & \xrightarrow{\quad} & M \end{array}$$

- Any short exact sequence

$$0 \rightarrow M \rightarrow N \rightarrow P \rightarrow 0$$

splits.

- The (covariant) functor $\text{Hom}_R(P, -)$ is exact (note left-exactness is always true for Hom functors)
- P is a direct summand of a free module⁶

Remark 3.1. Any free module is projective. Furthermore, the converse is occasionally (but not always!) true: a projective module is always free over a PID (basic fact), and a projective module over a local ring is always free (straightforward to prove for finitely generated modules via Nakayama's lemma from commutative algebra, but a difficult fact in general—keep the general result in mind only as a restriction for coming up with counterexamples).

Definition 3.7. An R -module, F , is said to be **flat** if any of the following equivalent conditions are true:

- The functor $F \otimes_R - \cong - \otimes_R F$ is exact (note right-exactness is always true for tensoring)
- The functor $F \otimes_R -$ takes injective maps to injective maps.
- (equational criterion for flatness) [“relations in F arise from relations in R ”] For any R -module homomorphism $\varphi : R^n \rightarrow F$, and any K a finitely generated submodule of $\ker \varphi$, φ factors through a map $\psi : R^n \rightarrow F'$ for some free⁷ R -module F' so that $K \subset \ker \psi$, e.g. such that the following diagram commutes:

$$\begin{array}{ccccc} & & & & F' \\ & & & \nearrow \text{dashed } \exists \psi & \searrow \text{dashed} \\ K & \xrightarrow{f.g.} & \ker \varphi & \hookrightarrow & R^n & \xrightarrow{\varphi} & F \\ & \searrow \text{dashed} & \swarrow \text{dashed} & & & & \\ & & \ker \psi & & & & \end{array}$$

- (translating the above into a concrete statement about relations—thank you wikipedia) For any R -linear relation

$$\sum_{i=1}^n r_i x_i = 0$$

with $r_i \in R$, $m_i \in F$, there exist some (“change of basis/generator”) elements $y_j \in F, a_{ij} \in R$ such that

$$\forall j, \sum_{i=1}^n r_i a_{ij} = 0$$

⁶Note: not every abelian category has a good notion of “free object”, so this particular description is distinct for categories of modules.

⁷Note: not every abelian category has a good notion of “free object”, so this particular description is distinct for categories of modules.

and

$$\forall i, x_i = \sum_{j=1}^m a_{ij} y_j.$$

Remark 3.2. Any projective module is necessarily flat. The converse is (“clearly”) true for finitely presented flat modules (hence for finitely generated flat modules over Noetherian rings).

There is yet another equivalent condition (which is, to my understanding, never taken as a definition):

Theorem 3.1. M is a flat R module if and only if, for every (finitely generated) ideal I of R , $M \otimes I \rightarrow M \otimes R$ is injective.

For a particularly elementary (if long) proof, see Dummit and Foote exercise 10.5.25 (more or less– an R). When we define (left)-derived functors, we will have another fancier restatement of the above available to us:

Theorem 3.2 (ideal-theoretic criterion for flatness). M is a flat R -module if and only if

$$\mathrm{Tor}_1^R(M, R/I) = 0$$

for all (finitely generated) ideals I of R .

Definition 3.8. An R -module, I , is said to be **injective** if any of the following equivalent statements are true:

- For any module homomorphism $M \rightarrow I$, any injective module homomorphism $M \rightarrow N$ admits an extension $N \rightarrow I$ (“along $M \rightarrow I$ ”) such that the following diagram commutes:

$$\begin{array}{ccc} I & \xleftarrow{\quad} & M \\ & \swarrow \text{dashed} & \downarrow \\ & & N \end{array}$$

- Any short exact sequence

$$0 \rightarrow I \rightarrow M \rightarrow N \rightarrow 0$$

splits.

- The (contravariant) functor $\mathrm{Hom}_R(-, I)$ is exact (note left-exactness [interpreted appropriately for variance] is always true for Hom functors)

Note that projective modules are “dual” to injective modules and vice versa. Here is a useful criterion for checking injectivity:

Theorem 3.3. (Baer’s Criterion) Let I be an R -module. Then, I is injective if and only if any homomorphism $J \rightarrow I$ from J a submodule/ideal of R admits an extension to a homomorphism $R \rightarrow I$.

As an immediate corollary, we have the following:

Corollary 3.1. Any abelian group A is injective (as a $\mathbb{Z}$ -module) if and only if it is divisible (e.g., the map $a \mapsto na$ is a surjection for all non-zero $n \in \mathbb{Z}$).

3.1 Ext and Tor

Now that we know that Tor is, we can provide a quick proof of theorem 3.2:

Proof of theorem 3.2. The forward implication is immediate (from Tor being balanced). For the reverse implication, We follow 24.3.4 in Vakil's FOAG.

Note that M is flat if and only if $\text{Tor}_1^R(M, N) \cong \text{Tor}_1^R(N, M) = 0$ for all N (in which case the LES of Tor gives a SES).

Now, if N is finitely generated take $\{r_1, \dots, r_n\}$ a set of generators. We have $A/\text{Ann}(r_n) \cong Ar_n$ a submodule of N , hence we have a SES

$$0 \rightarrow A/\text{Ann}(r_n) \rightarrow N \rightarrow N/\langle r_n \rangle \rightarrow 0$$

a SES where $N/\langle r_n \rangle$ is generated by $\langle r_1, \dots, r_{n-1} \rangle$. Taking Tor and inducting on n we get the desired result for finitely generated modules, since the base case $n = 1$ gives $N \cong R/\text{Ann}(r_1)$.

In general, a module N is a union of its finitely generated submodules, hence a *filtered colimit* (the filtering given by set-theoretic inclusion) of its f.g. submodules. By general abstract nonsense, filtered colimits are exact functors (exercise 1.5.L of Vakil) hence commute with homology by theorem 4.2, hence $\text{Tor}_1^R(M, N) = 0$. $\square$

4 Generalities on “Abelian Categories”

An “abelian category” can, in some sense, be thought of as a “convenient category to do homological algebra in” (which may not *literally* be a category of R -modules over a ring R). There are a number of different equivalent ways to define/build up the definition of an abelian category, but ultimately what it comes down to is as follows: an abelian category should

- Have a way for us to (compatibly) add morphisms in the same Hom (that is, $\text{Hom}(a, a')$ has the structure of an abelian group in a nice way)
 - Ideally, it should be possible to imagine maps in terms of matrices (e.g., like we can with vector spaces) and composition in terms of matrix multiplication
- Should have some “0 object” (e.g., like a 0-module or trivial group)
- should have products and sums (that is, coproducts)
 - In addition, they should agree in the finite case.
- Should let us *always* be able to take kernels and cokernels.
 - Furthermore, unlike in **Group** where we need to distinguish “normal subgroups” from every subgroup, we’d like *every* subobject to be something we can quotient by– hence, every not-quite-injection (monomorphism) should be a kernel (e.g., of *something*).
 - Dually, we’d like every surjection to be a quotient map– hence, every not-quite-surjection (epimorphism) should be a cokernel (e.g., of *something*)
- Should have the isomorphism theorem(s) hold– e.g., for $\varphi : M \rightarrow N$, we should always have $\text{im } \varphi \cong M / \ker \varphi$.⁸

We can keep on going and listing nice properties of abelian/module categories, but for now let’s leave it here. As it happens, various subsets of these conditions give the rest⁹. So, we just choose some subset of these properties, build these up in steps, and then use that as the “definition” (noting that the rest of the things we want will hold). Picking our poison, we begin:

Definition 4.1. A category, $\mathcal{C}$, is said to be **Ab-enriched** or **preadditive** if, for every pair of $\mathcal{C}$ -objects M, N , $\text{Hom}_{\mathcal{C}}(M, N)$ has the structure of an abelian group and composition is $\mathbb{Z}$ -bilinear– e.g.,

$$\varphi \circ (n\psi_1 + m\psi_2) = n\varphi \circ \psi_1 + m\varphi \circ \psi_2$$

and

$$(n\psi_1 + m\psi_2) \circ \varphi = n\psi_1 \circ \varphi + m\psi_2 \circ \varphi.$$

You’ll note that being $\mathbb{Z}$ -bilinear (“biadditive”, if you prefer) is more or less just left and right distributivity. This should be familiar to you: recall that for modules, the group of endomorphisms of a module form a *ring* (e.g., the endomorphisms of a finite-dimensional k -vectorspace are a ring of matrices), etc; more generally, we like to think of “composing linear transformations” like multiplying matrices (even between vector spaces of different dimension), so requiring “biadditivity” is really just a way to ensure this analogy still works.¹⁰ Next up, we say:

⁸Note that the left handside is $\ker \text{coker } \varphi$ and the right-hand-side is $\text{coker } \ker \varphi$, sometimes called the “coimage”– this can be phrased diagrammatically if you like (as Jason did in class). More generally, one has a natural morphism $\lim \text{colim} \rightarrow \text{colim } \lim$, so the isomorphism theorem can be stated as “the natural morphism $\ker \text{coker} \rightarrow \text{coker } \ker$ is an isomorphism.”

⁹indeed– as a neat curiosity, if your category has “binary biproducts”, all kernels and cokernels, and a 0 object, then that gives an abelian group structure on the Hom sets. If you’d like, then, you can consider **Ab**-enrichment as a *property* of a category and not an extra structure on said category, but I’ve no sense for how this would be a *useful* consideration.

¹⁰One can formally prove that morphisms can be treated in terms of matrices *so long as* the category in question has finite biproducts

Definition 4.2. An $\mathbf{Ab}$ -enriched/preadditive category, $\mathcal{C}$, is said to be additive if it has a 0 object¹¹ and has all binary/finite products.

You’ll note we didn’t actually say anything about coproducts existing, nor about finite products and coproducts coinciding. You can check this yourself (or just go to the corresponding nlab page), but indeed: if we have an appropriate abelian group structure on all homs, then every product is a coproduct and vice versa, so we only need to specify the existence of one or the other.

Definition 4.3. An additive category, $\mathcal{A}$, is said to be abelian if:

- Every morphism $\varphi : M \rightarrow N$ has a kernel and cokernel.¹²
- Every monomorphism and epimorphism is “normal”—that is, every monomorphism is the kernel of something, and every epimorphism is the cokernel of something.

In summary, a category is abelian if: it is $\mathbf{Ab}$ -enriched, has 0 object, all finite products, kernels, and cokernels, and if every monomorphism is a kernel and every epimorphism is a cokernel. This will end up being sufficient to imply all the other properties that we’d want/like (including the isomorphism theorems above)

Remark 4.1. Jason might require us to technically be able to unravel the definition, but, in practice “abelian category” just means “category which we can think of like a category of left/right R -modules”. This is possible to formalize by the 4.1 Freyd-Mitchell Embedding Theorem, which is technically useful. It’s also abstract nonsense, and for most things we talk about, you can use the words “abelian category” interchangeably with “category of R -modules” (or, if you really want, “category of chain complexes of R -modules”), so just think in terms of whatever’s easiest for now (usually, it’ll be clear when one’s research/area of interest is such that the notion of “abelian category” is useful.)

Theorem 4.1. (Freyd-Mitchell Embedding) Let $\mathcal{A}$ be a (small) abelian category. Then, there is a ring R (not necessarily commutative) and a full, faithful, exact (hence additive) functor $i : \mathcal{A} \otimes_R \mathbf{Mod}$ the category of left R -modules.

You do *not* need to memorize this (nor will we see a proof), but I feel it necessary to mention it as a name to look up/be referred to.

One last thing for us to define in generality: we defined additive functors before in module categories, but we may as well just define it again in general:

Definition 4.4. A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ between pre-additive categories is said to be **additive** if, for all a, a' in $\mathcal{A}$, we have that the induced map $\mathrm{Hom}_{\mathcal{A}}(a, a') \rightarrow \mathrm{Hom}_{\mathcal{B}}(Fa, Fa')$ is an abelian group homomorphism.¹³ We also have definitions for (left, right, half) exact functors and injective/projective objects—these are more or less verbatim the definitions for categories of modules, but we don’t have any convenient characterizations in terms of free objects or ideals. In that section, we had to restate various definitions etc. “for each side” because the opposite of a module category need not be a module category. However, if you look at the axioms we stated for an abelian category, you’ll note that almost all of them are self-dual! In particular, we see

Proposition 4.1. If $\mathcal{A}$ is an abelian category, then its opposite $\mathcal{A}^{op}$ is an abelian category.

This sort of justifies the way we treated, say, left-right exactness of contravariant functors etc. In particular, since kernels in $\mathcal{A}$ are precisely cokernels in $\mathcal{A}^{op}$ (and vice versa), “ $F : \mathcal{A}^{op} \rightarrow \mathcal{B}$ preserving (co)kernels” is precisely the statement we wrote out before.

¹¹A 0 object is an object that is both initial and terminal. In particular, for any M, N , we have a map $M \rightarrow 0 \rightarrow N$ —the “zero map”. As the terminology suggests, this is precisely the “zero homomorphism” of groups/modules, and then all the things we’d like to be true for a “zero map” just end up following formally from the fact it factors through 0.

¹²If $\mathcal{A}$ satisfies this axiom, but not the 2nd, it is sometimes called “pre-abelian”—for example, the category of Banach spaces with bounded/continuous linear maps is pre-abelian, but fails that last axiom—e.g., the inclusion $\ell^1(\mathbb{R}) \rightarrow c_0(\mathbb{R})$ is monic and epic but is neither a kernel nor cokernel [according to [this mathstackexchange thread](#)]. Anyways, neat concept and all but I don’t really want to add yet another definition, and it’s not like we really need/care about pre-abelian categories here anyways.

¹³If you’re absurd (like nlab), then you can equivalently define an additive functor between additive categories as one which preserves finite “biproducts”—this more or less follows since (as mentioned in a prior footnote) the abelian group structure on Hom sets can be “derived” from the finite biproducts.

4.1 (Co)homology on abelian categories

Definition 4.5. A δ -*functor*

Definition 4.6. A δ -functor, H^* *universal* if

Theorem 4.2. (FOAG 1.5.I, “the FHHF theorem”) Suppose $F : \mathcal{A} \rightarrow \mathcal{B}$ is an additive functor of abelian categories, and $C^\bullet$ an $\mathcal{A}$ -complex.

- If F is right exact, there exists a natural morphism $FH^\bullet \rightarrow H^\bullet F$
- If F is left exact, there is a natural morphism $H^\bullet F \rightarrow FH^\bullet$ (e.g., $FH^\bullet \leftarrow H^\bullet F$)
- If F is exact, then the above are inverse and there is a natural isomorphism $FH^\bullet \xrightarrow{\sim} H^\bullet F$.

Proof.

□

This is abstract nonsense, but generally quite useful abstract nonsense (in that it gives very quick/easy proofs, e.g. that localization commutes with homology etc).¹⁴

¹⁴For those so inclined, one can google the definition of “filtered category” (a categorical generalization of “poset of set-theoretic inclusions of subsets”) to make sense of a “filtered colimit”- Then, putting in a bit of work, one can show that taking filtered colimits gives an exact functor *on module categories*. (This is not true in general, cf. Grothendieck’s AB5 axiom for “nice(r) abelian categories.”)